

INTRODUCTION

With the advent of autonomous vehicles camera-based navigation is increasingly being used. A classic solution is to employ visual odometry with monochrome cameras.¹ An alternative method is to use UV polarized photodiodes. This technique is inspired by nature. Many animals, such as the field cricket, use skylight polarization for navigation.²

Thanks to the emergence of new low-cost technology like Monochrome Polarization Filter Arrays (MPFA) with the Sony IMX250 MZR (see Figure 1),³ it is now possible to capture both of spectral and polarization information in a single shot way. Mixing spectral and polarization imaging is therefore a new kind of method for navigation.

The aim of this study is to develop a new low-cost navigation solution based on MPFA cameras and on the skylight polarization.

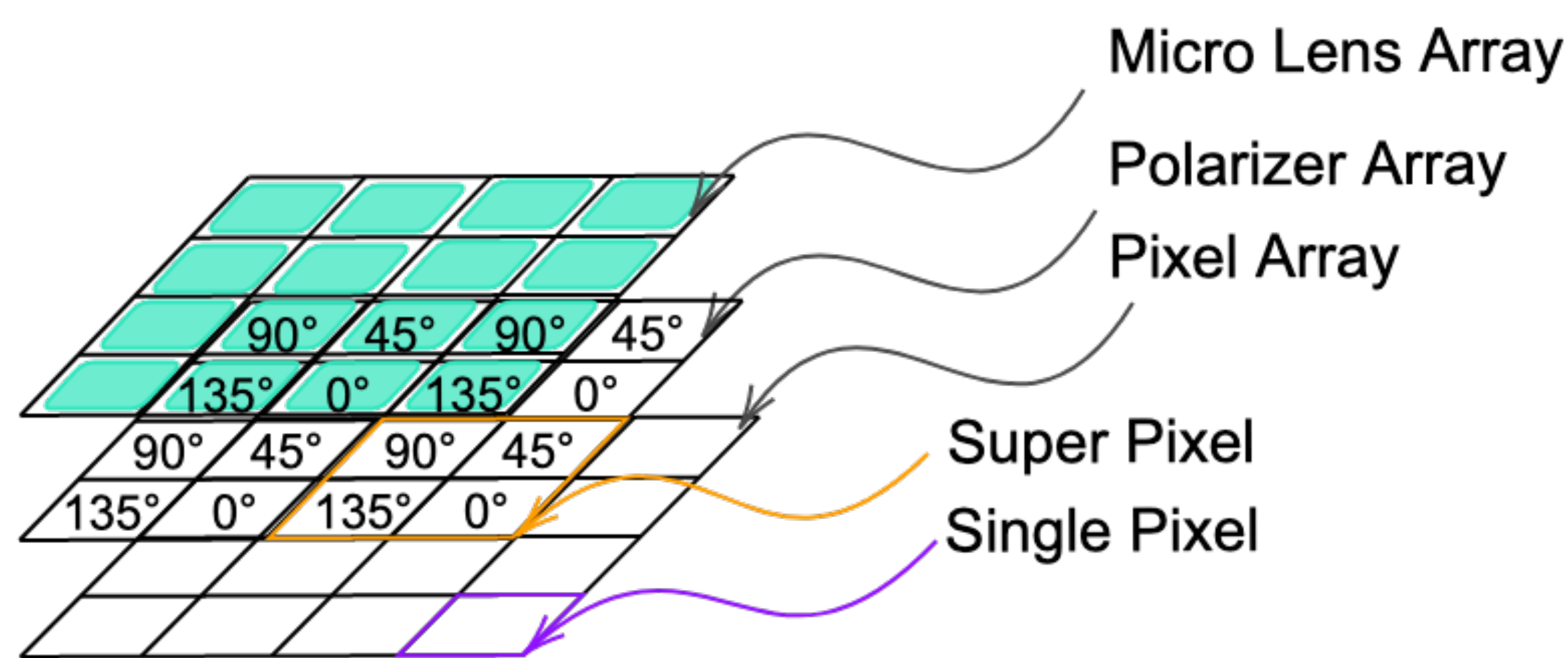


Figure 1: MPFA polarimeter architecture. Here is the sensor arrangement specific to the Sony IMX250 MZR.

MECHANICS

The advantages of this camera technology are:

1. No moving part.
2. No need to use several cameras and therefore to realign the images.
3. Low-cost.

We focused on two main objectives:

1. Capture data on both static and dynamic platforms.
2. Evaluate robustly the navigation algorithm using redundant sources of reference data.

We have therefore built two mechanical platforms:

1. Rotary (for benchmarking and performance evaluation in static configuration, see Figure 2a).
2. Embedded (for benchmarking and validation under dynamic scenes, see Figure 2b).

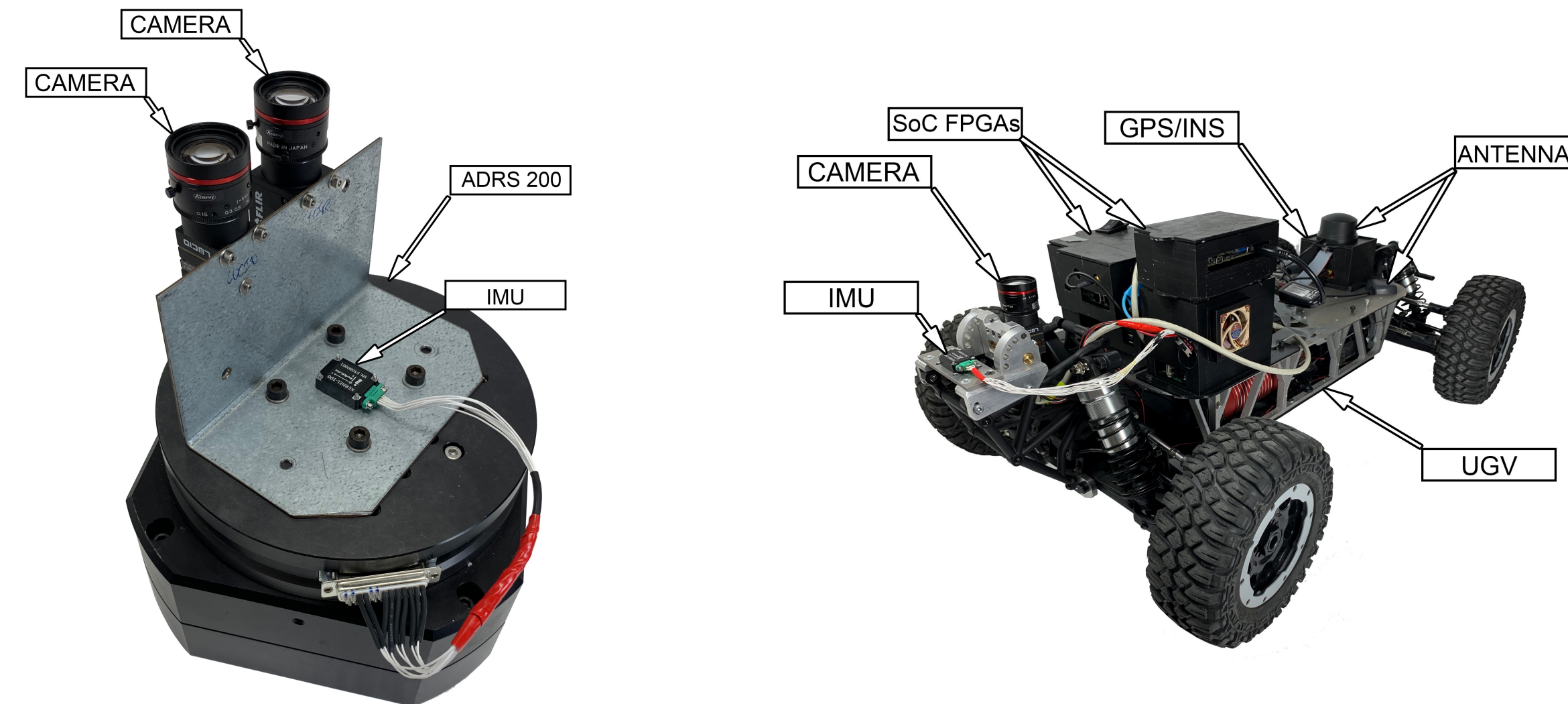


Figure 2: (a) The rotary setup and (b), the embedded setup.

NAVIGATION

The navigation algorithm is partly based on the literature and on the Rayleigh scattering model (see Figure 3). A threshold-extraction method was used to identify the points whose values are around $\phi = 0^\circ / 180^\circ$ in the Angle of Linear Polarization that characterized the direction of the sun (see Figure 3). In the paper from Lu *et al.*⁴ the sun direction was assessed using a Hough transform. We have proposed to use a maximum likelihood estimation method in order to reduce the computation time. Figure 4 shows the processing pipeline.

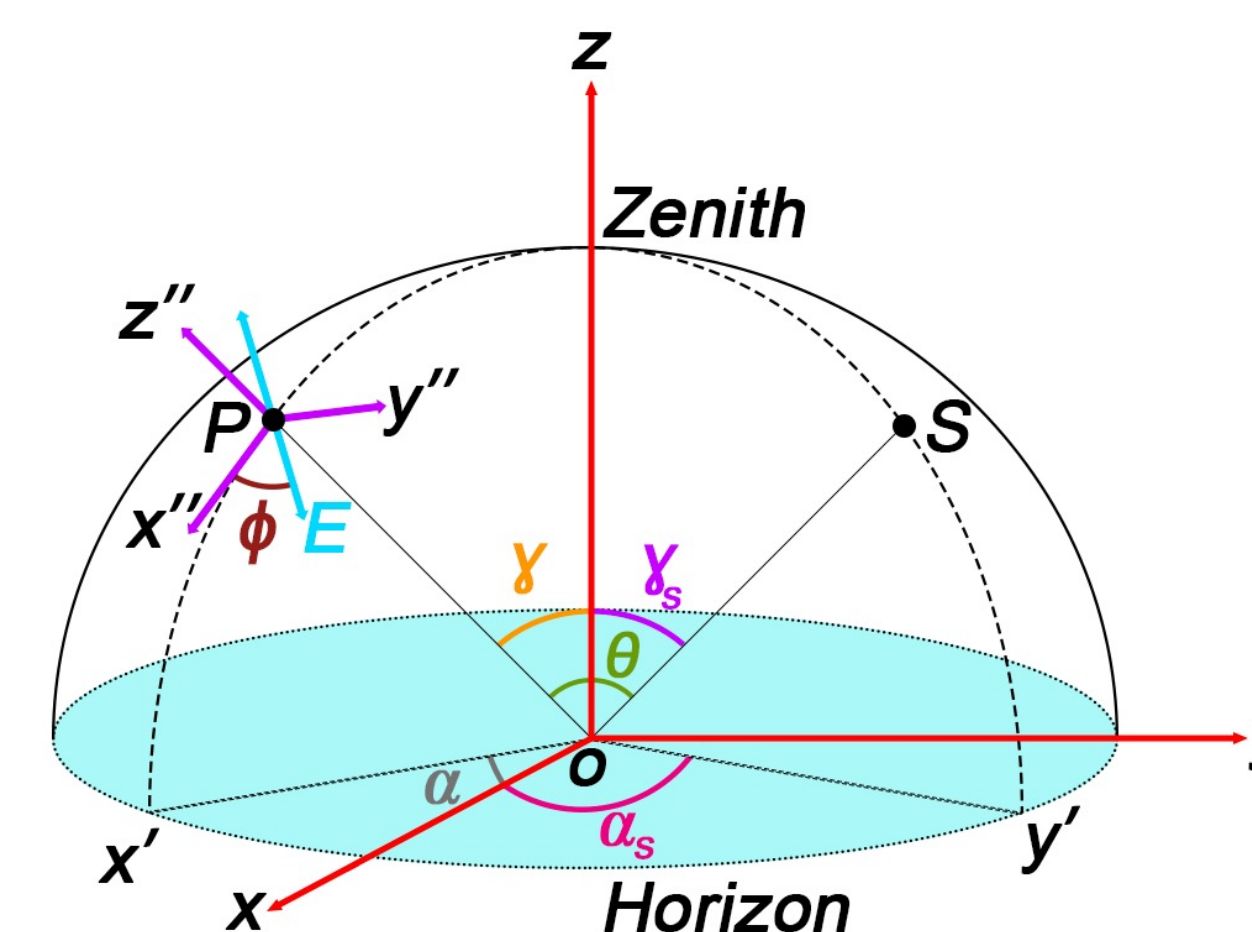


Figure 3: Skylight polarization: E-vector direction in a local frame with γ : zenith angle, α : azimuth angle, ϕ : polarization angle, and θ : scattering angle. S belongs to the Sun.

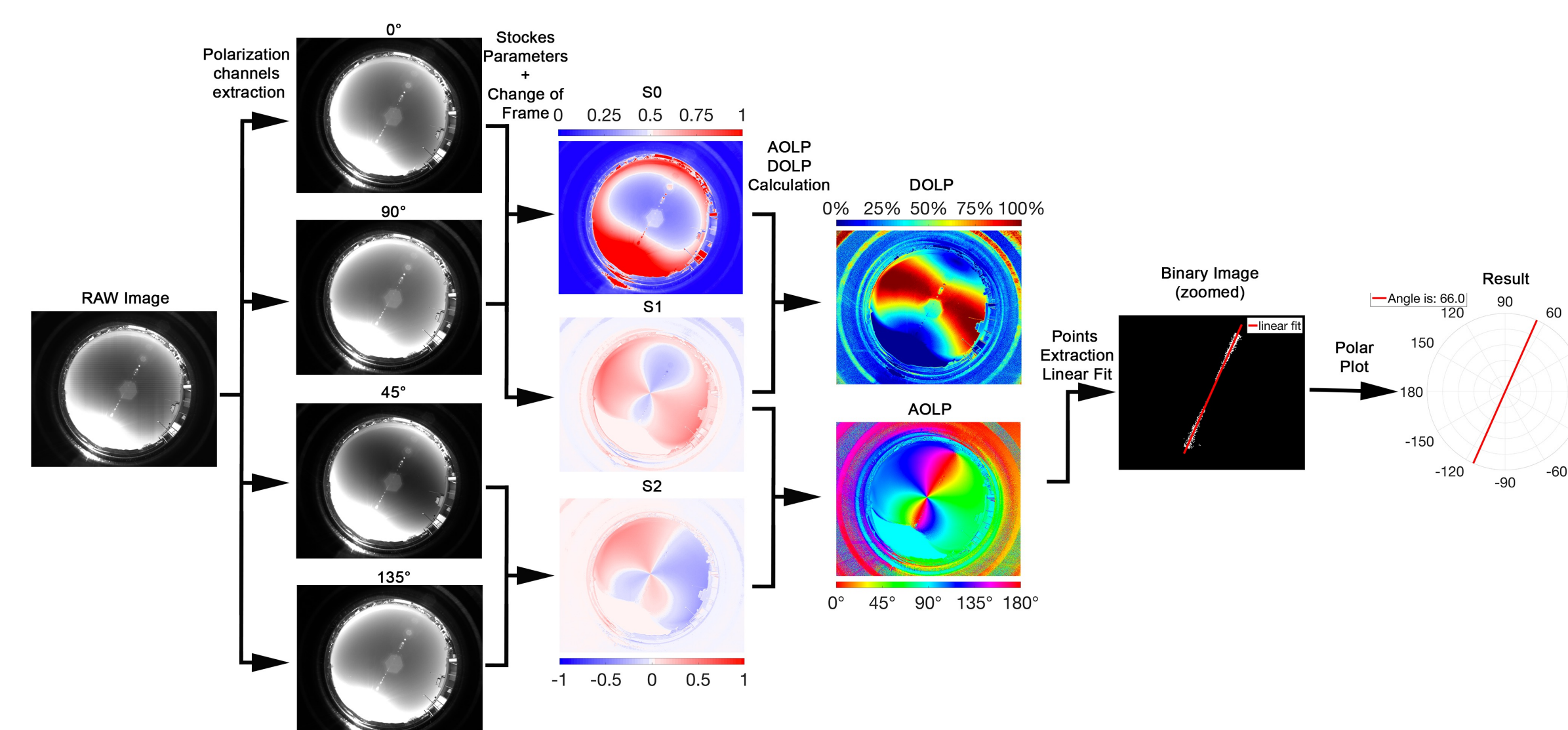


Figure 4: Processing steps to estimate the solar meridian (for illustration purpose, images have been taken with a fisheye lens. All Stokes and polarimetric images are already shown in the meridian frame).

RESULTS

Figure 5 presents a comparison between the heading given by the FLIR camera and that given by the GPS-IMU system of the UGV. This allows us to compare on field data.

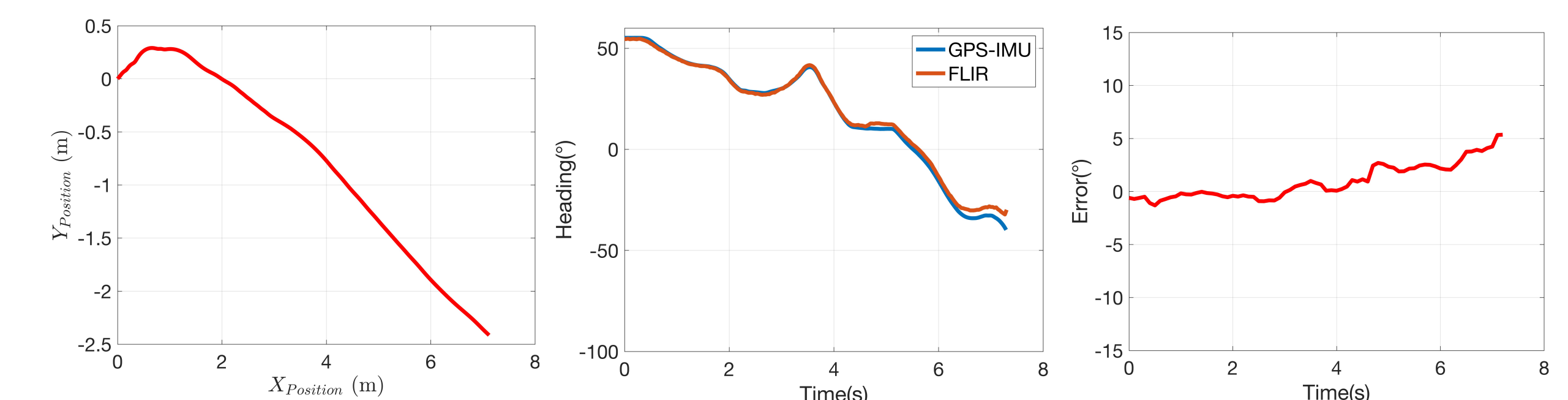


Figure 5: UGV Heading comparison between GPS-IMU and FLIR Camera (acquisition campaign of 2021/03/30 at 2:58pm, GPS-IMU with 100Hz rate, Camera with 10Hz rate): (a) UGV trajectory, (b) Heading comparison, (c) Fitting error.

We can see that globally the curves have the same trend but are not exactly superimposed.

This can be partly due to the noise in the image (e.g. saturation), atmospheric conditions (e.g. pollution), levelling of the UGV, or geometrical lens aberrations.

PERSPECTIVES

The perspectives of this work are:

1. To improve accuracy and robustness.
2. To implement the algorithm in real time.
3. To investigate the spectral band impact.

In conclusion, this poster shows a new navigation system based on the skylight polarization.

As a future work, the solution could be modified to be integrated into a higher dynamic vehicle, such as a unmanned aerial vehicle (UAV).

REFERENCES

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3. Sony, "Polarsens polarization image sensor" Technical Report (2018).
4. Lu, H., et al., Angle algorithm based on Hough transform for imaging polarization navigation sensor. *Optics express*, 2015. 23(6): p. 7248-7262.